

▼ HW #1: Decision Making and Loops

▼ Problem #1.1 : add the integers 3 and 12.

```
#Vikram Nadathur May 26th 2026  
3 + 12
```

```
15
```

▼ Problem #1.2 : Subtract 100.5 from 250.75.

```
#Vikram Nadathur May 26th 2026  
250.75 - 100.5
```

```
150.25
```

▼ Problem #1.3 : multiply 15.0 and 10.0.

```
#Vikram Nadathur May 26th 2026  
15.0 * 10.0
```

```
150.0
```

▼ Problem #1.4 : divide 500 by 20.

```
#Vikram Nadathur May 26th 2026  
500/20
```

```
25.0
```

▼ Problem #1.5 : add 13.4 and 27.6.

```
#Vikram Nadathur May 26th 2026  
13.4+27.6
```

```
41.0
```

▼ Problem #1.6 : multiply 1000.5 and 3000.

```
#Vikram Nadathur May 26th 2026  
1000.5*3000
```

```
3001500.0
```

▼ Problem #1.7 : divide 500.5 by 25.

```
#Vikram Nadathur May 26th 2026  
500.5/25
```

```
20.02
```

▼ Problem #2.1 : let's create a variable called gpt_version and assign a string to it with the current version of GPT you are using.

NOTE: We will output the variable's value by writing the name of the variable on a line of its own.

```
gpt_version = "GPT-3"  
gpt_version
```

```
'GPT-3'
```

- ✓ **Problem #2.2 :** let's create and name a variable with the number of parameters in GPT-3 as an integer (175 billion). Output its value.

```
#Vikram Nadathur May 26th 2026
params = 175000000000
params
```

```
175000000000
```

- ✓ **Problem #2.3:** look through the given declarations and determine which ones are incorrect.

- a. model_version = 3.5
- b. Model4 = "GPT-4"
- c. 2model = "GPT-2"
- d. first_last_name = "john_doe"
- e. Hello! = 0.283
- f. variable_24 = "OpenAI"
- g. str = "string"

Double Click This Cell To Enter Your Answer(s)

c,e

- ✓ **Problem #3.1:** let's use a for loop to print the numbers from 1 to 5.

```
#Vikram Nadathur May 26th 2026
for i in range(1, 6):
    print(i)
```

```
1
2
3
4
5
```

- ✓ **Problem #3.2:** let's use a while loop to print the numbers from 5 to 1.

```
#Vikram Nadathur May 26th 2026
i = 5
while i > 0:
    print(i)
    i=i-1
```

```
5
4
3
2
1
```

- ✓ **Problem #3.3:** let's create a list of integers and use a for loop to find the sum of all the numbers in the list.

```
#Vikram Nadathur May 26th 2026
numbers = [1, 2, 3, 4, 5]
total = 0
for number in numbers:
    total += number
print(total)
```

```
15
```

- ✓ **Problem #3.4:** use a for loop to print each character in the string "ChatGPT".

```
#Vikram Nadathur May 26th 2026
string = "ChatGPT"
for char in string:
    print(char)
```

```
C
h
a
t
G
P
T
```

- ▼ **Problem #3.5:** use a while loop to print the numbers from 10 down to 1.

```
#Vikram Nadathur May 26th 2026
i=10
while i > 0:
    print(i)
    i=i-1
```

```
10
9
8
7
6
5
4
3
2
1
```

- ▼ **Problem #3.6:** let's define a function called greet that takes a name as an argument and returns a greeting message.

```
#Vikram Nadathur May 26th 2026
def greet(name):
    return "Hello, " + name
```

- ▼ **Problem #3.7:** let's define a function called add_numbers that takes two numbers as arguments and returns their sum.

```
#Vikram Nadathur May 26th 2026
def add_numbers(a, b):
    return a + b
```

- ▼ **Problem #3.8:** let's define a function called is_even that takes a number as an argument and returns True if the number is even, and False otherwise.

```
#Vikram Nadathur May 26th 2026
def is_even(number):
    return number % 2 == 0
```

- ▼ **Problem #3.9:** let's define a function called multiply_numbers that takes two numbers as arguments and returns their product. *italicized text*

```
#Vikram Nadathur May 26th 2026
def multiply_numbers(a, b):
    return a * b
```

- ▼ **Problem #3.10:** let's define a function called count_vowels that takes a string as an argument and returns the number of vowels in the string.

```
#Vikram Nadathur May 26th 2026
def count_vowels(string):
    vowels = "aeiouAEIOU"
    count = 0
    for char in string:
```

```
    if char in vowels:
        count += 1
    return count
```

Below is the examples with a list/array in Python

```
numbers = [1, 2, 3, 4, 5]
numbers.append(6) #6 is added
numbers.remove(2) #2 was removed
print(numbers)
```

```
[1, 3, 4, 5, 6]
```

- ✓ **Problem #4.1.** : let's define a function called `find_max` that takes a list of numbers as an argument and returns the largest number in the list using a for loop.

```
def find_max(numbers):
    max_number = numbers[0]
    for number in numbers:
        if number > max_number:
            max_number = number
    return max_number

# Test the function
numbers = [1, 5, 3, 9, 2]
print(find_max(numbers)) # Should print 9
```

```
9
```

- ✓ **Problem #4.2** : let's define a function called `find_min` that takes a list of numbers as an argument and returns the smallest number in the list using a for loop.

```
#Vikram Nadathur May 26th 2026
def find_min(numbers):
    for number in numbers:
        minnum=numbers[0]
        if number < minnum:
            minnum = number
    return minnum

# Test the function
numbers = [1, 5, 3, 9, 2]
print(find_min(numbers)) # Should print 1
```

```
1
```

- ✓ **Problem #4.3:** let's define a function called `calculate_average` that takes a list of numbers as an argument and returns the average of the numbers.

```
#Vikram Nadathur May 26th 2026
def calculate_average(numbers):
    return sum(numbers) / len(numbers)

# Test the function
numbers = [1, 5, 3, 9, 2]
print(calculate_average(numbers)) # Should print 4.0
```

```
4.0
```

- ✓ **Problem #4.4:** define a function called `count_even_numbers` that takes a list of numbers as an argument and returns the count of even numbers in the list.

```
#Vikram Nadathur May 26th 2026
def count_even_numbers(numbers):
    count = 0
    for number in numbers:
```

```
if number % 2 == 0:
    count += 1
return count
```

- ✓ **Problem #4.5:** define a function called `count_odd_numbers` that takes a list of numbers as an argument and returns the count of odd numbers in the list.

```
#Vikram Nadathur May 26th 2026
def count_odd_numbers(numbers):
    count = 0
    for number in numbers:
        if number % 2 != 0:
            count += 1
    return count
```

- ✓ **Problem #4.6:** define a function called `find_sum_of_squares` that takes a list of numbers as an argument and returns the sum of the squares of the numbers.

```
#Vikram Nadathur May 26th 2026
def find_sum_of_squares(numbers):
    total = 0
    for number in numbers:
        total += number ** 2
    return total
```

- ✓ **Problem #4.7 :** let's create a function called `check_positive` that takes a number as an argument and returns `True` if the number is positive, and `False` otherwise.

```
def check_positive(number):
    return number > 0

# Test the function
print(check_positive(5)) # Should print True
print(check_positive(-3)) # Should print False
```

```
True
False
```

- ✓ **Problem #4.8:** create a function called `check_negative` that takes a number as an argument and returns `True` if the number is negative, and `False` otherwise.

```
#Vikram Nadathur May 26th 2026
def check_negative(number):
    if(number>=0):
        return True
    else:
        return False

# Test the function
print(check_negative(-5)) # Should print True
print(check_negative(3)) # Should print False
```

```
False
True
```

- ✓ **Problem #4.9:** create a function called `check_divisible` that takes two numbers as arguments and returns `True` if the first number is divisible by the second, and `False` otherwise.

```
#Vikram Nadathur May 26th 2026
def check_dvisible(num1,num2):
    if num1%num2==0:
        return True
    else:
        return False
```

End of HW #1

© 2024 CPS*3320, Kean University